

Lecture-29
Mathematics 1 (15B11MA111)

CO [C105.5]

Module: Laplace Transforms

Topics to be covered

- ❖ Inverse Laplace transform.
- ❖ Evaluation of Integrals by Laplace transform

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Inverse Laplace Transform

If the Laplace transform of a function $f(t)$ is $F(s)$ then $f(t)$ is **Inverse Laplace Transform** of $F(s)$.

Symbolically, if $L\{f(t)\} = F(s)$ then $f(t) = L^{-1}\{F(s)\}$

The inverse Laplace transform of $F(s)$ is given by the complex integral

$$f(t) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} F(s) e^{st} ds$$

However in most of cases we can find inverse Laplace transform by simplifying and rearranging the function $F(s)$ in terms of functions whose inverse transforms are available in tables.

Example 1: Find $f(t) = L^{-1}\{F(s)\}$ given that

$$F(s) = \frac{(s+5)(s+3)}{s(s+2)(s+6)}$$

Solution: Let

$$F(s) = \frac{c_1}{s} + \frac{c_2}{s+2} + \frac{c_3}{s+6}$$

We determine each constant by setting s equal to the root from the corresponding denominator

$$C_1 = \left. \frac{(s+5)(s+3)}{(s+2)(s+6)} \right|_{s=0} = \frac{(5)(3)}{(2)(6)} = \frac{5}{4}$$

$$C_2 = \left. \frac{(s+5)(s+3)}{s(s+6)} \right|_{s=-2} = \frac{(3)(1)}{(-2)(4)} = -\frac{3}{8}$$

$$C_3 = \left. \frac{(s+5)(s+3)}{s(s+2)} \right|_{s=-6} = \frac{(-1)(-3)}{(-6)(-4)} = \frac{1}{8}$$

$$\text{Therefore } F(s) = \frac{5}{4} \frac{1}{s} - \frac{3}{8} \frac{1}{s+2} + \frac{1}{8} \frac{1}{s+6}$$

$$\text{We have } L\{1\} = \frac{1}{s} \quad L\{e^{at}\} = \frac{1}{s-a} \quad \therefore f(t) = \frac{5}{4} - \frac{3}{8}e^{-2t} + \frac{1}{8}e^{-6t}$$

Example 2 Find $f(t) = L^{-1}\{F(s)\}$ given that $F(s) = \frac{4s+16}{s^2+4s+5}$

Solution: *First Approach*

$$F(s) = \frac{4s+16}{s^2+4s+5} = \frac{4s+16}{(s+2+i)(s+2-i)} = \frac{A}{s+2+i} + \frac{B}{s+2-i}$$

$$A = \left. \frac{4s+16}{s+2-i} \right|_{s=-2-i} = \frac{4(-2-i)+16}{-2-i+2-i} = \frac{8-4i}{-2i} = 4i+2$$

$$B = \left. \frac{4s+16}{s+2+i} \right|_{s=-2+i} = \frac{4(-2+i)+16}{-2+i+2+i} = \frac{8+4i}{2i} = 2-4i$$

$$\therefore F(s) = \frac{2+4i}{s+2+i} + \frac{2-4i}{s+2-i} \quad \because L\{e^{at}\} = \frac{1}{s-a}$$

$$\begin{aligned} \therefore f(t) &= (2+4i)e^{-(2+i)t} + (2-4i)e^{-(2-i)t} \\ &= e^{-2t}[(2+4i)e^{-it} + (2-4i)e^{it}] \\ &= e^{-2t}[2(e^{-it} + e^{it}) - 4i(-e^{-it} + e^{it})] = e^{-2t}[4\cos t + 8\sin t] \end{aligned}$$

Second Approach

$$F(s) = \frac{4s+16}{s^2+4s+5} = \frac{4s+16}{s^2+4s+4+1} = \frac{4s+16}{(s+2)^2+1}$$

We know that $L\{e^{-at} \sin \omega t\} = \frac{\omega}{(s+a)^2 + \omega^2}$ and

$$L\{e^{-at} \cos \omega t\} = \frac{s+a}{(s+a)^2 + \omega^2}$$

Since denominator is already in this form, we rearrange the numerator in $F(s)$ also and get

$$\begin{aligned} F(s) &= \frac{4s+16}{(s+2)^2+1} = \frac{4(s+2)+8}{(s+2)^2+1} \\ &= 4 \frac{s+2}{(s+2)^2+1} + 8 \frac{1}{(s+2)^2+1} \end{aligned}$$

$$\therefore f(t) = e^{-2t} [4 \cos t + 8 \sin t]$$

Some important results used for inverse Laplace transforms

$$\text{If } L\{f(t)\} = F(s), \text{ then } L^{-1}\{F(s+a)\} = e^{-at} L^{-1}\{F(s)\} = e^{-at} f(t)$$

$$\text{If } L\{f(t)\} = F(s) \text{ and } f(0) = 0, \text{ then } L^{-1}\{sF(s)\} = \frac{d}{dt} f(t)$$

$$\text{If } L\{f(t)\} = F(s), \text{ then } L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(u) du$$

$$\text{If } L\{f(t)\} = F(s), \text{ then } L^{-1}\left\{-\frac{d}{ds} F(s)\right\} = tf(t)$$

Convolution Theorem

$$\text{If } L\{f(t)\} = F(s) \text{ and } L\{g(t)\} = G(s), \text{ then}$$

$$L^{-1}\{F(s)G(s)\} = \int_0^t f(u)g(t-u)du$$

Example: Find the inverse Laplace transform of $\frac{1}{s(s+a)^2}$

Solution: We have

$$L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(u)du$$

$$\therefore L^{-1}\left\{\frac{1}{s(s+a)^2}\right\} = \int_0^t L^{-1}\left\{\frac{1}{(s+a)^2}\right\} dt \quad \because L\{te^{at}\} = \frac{1}{(s-a)^2}$$

$$\begin{aligned}\therefore L^{-1}\left\{\frac{1}{s(s+a)^2}\right\} &= \int_0^t te^{-at} dt \\ &= \left\{t \frac{e^{-at}}{-a} - \frac{e^{-at}}{a^2}\right\}_0^t \\ &= -t \frac{e^{-at}}{a} - \frac{e^{-at}}{a^2} + \frac{1}{a^2}\end{aligned}$$

Example: Using convolution theorem find $L^{-1} \left\{ \frac{1}{s^2(s-a)} \right\}$.

Solution: We have $L^{-1} \left\{ \frac{1}{s^2} \right\} = t = f(t)$

$$L^{-1} \left\{ \frac{1}{(s-a)} \right\} = e^{at} = g(t)$$

Using convolution theorem

$$L^{-1} \{F(s)G(s)\} = \int_0^t f(u)g(t-u)du$$

$$\begin{aligned} L^{-1} \left\{ \frac{1}{s^2(s+a)} \right\} &= \int_0^t u \cdot e^{a(t-u)} du \\ &= e^{at} \int_0^t u \cdot e^{-au} du \\ &= e^{at} \left\{ u \frac{e^{-au}}{-a} - \frac{e^{-au}}{a^2} \right\}_0^t \\ &= e^{at} \left\{ -t \frac{e^{-at}}{a} - \frac{e^{-at}}{a^2} + \frac{1}{a^2} \right\} = \frac{e^{at}}{a^2} - \frac{t}{a} - \frac{1}{a^2} \end{aligned}$$

Evaluation of Integrals by Laplace transform

Evaluate the integral

$$\int_0^{\infty} t e^{-3t} \sin t dt$$

We have $L\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$

Therefore $\int_0^{\infty} t e^{-3t} \sin t dt = L\{t \sin t\} \Big|_{s=3}$

$$= (-1) \frac{d}{ds} L\{\sin t\} \Big|_{s=3} = (-1) \frac{d}{ds} \frac{1}{s^2 + 1} \Big|_{s=3}$$

$$= \frac{2s}{(s^2 + 1)^2} \Big|_{s=3} = \frac{3}{50}$$

Practice Questions:

1. Using convolution theorem find $L^{-1} \left\{ \frac{3}{(s^2+9)^2} \right\}$

Ans. $\frac{1}{18} [\sin 3t - 3t \cos 3t]$

2. Evaluate the integral $\int_0^\infty \frac{\sin t}{t} dt$

Ans. $\frac{\pi}{2}$

Thank you